

Global Ratio Monotonicity for a Killed von Mises Bridge

[COMPLETE-THEOREM MANUSCRIPT — SEAL IN PROGRESS; DO NOT
SUBMIT.]

Pending relay lemmas are marked [SLOT].]

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Abstract

For $\beta > 0$ let $I_m = I_m(\beta)$ denote modified Bessel functions of the first kind, and set $a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2]$, $b_m = mI_m^4$, $F_A(t) = \sum_{m \geq 1} a_m \sin(mt)$, $F_B(t) = \sum_{m \geq 1} b_m \sin(mt)$, and $E = F_A/(2F_B)$. The global ratio-monotonicity problem for the surface expansion of a two-dimensional $SU(2)$ lattice gauge observable is: (i) $F_B > 0$ on $(0, \pi)$, and (ii) $E' < 0$ on $(0, \pi)$, for every $\beta > 0$. We prove (i) twice, and prove (ii) for $0 < \beta \leq 3$ via an exact pair-identity skeleton plus an archived interval certificate executed by two independent interval-arithmetic implementations. The structural core is exact: E is, as an algebraic identity, the mean of $\cos \psi$ under the midpoint law of a four-step killed von Mises bridge; the generating kernels reduce, via the Neumann addition theorem, to two-dimensional integrals of a *single* Bessel function whose saddle deficit is an exact sum of two squares; and exact saddle cancellations yield the coefficients of the (verified) closed second-order law $E = \cos(t/2)(1 - c(t)/\beta) + O(\beta^{-2})$, $c(t) = (4 \cos^2(t/4) - 1)/(2 \cos(t/4) \cos(t/2))$. Three certified negative results (interval arithmetic, two implementations, nested enclosures) kill every monotone full-path coupling, with an exact mechanism at threshold $\beta |\cos t| = 3/2$. For the remaining range we give a two-scale closure map with a verified candidate threshold $\beta_0 = 15$ and, at the π -endpoint, exact identities for the cubic coefficient c_3 (telescoped alternating form, integral form, parity) together with its verified prefactor law. Every claim is labelled exact / certified / verified; the machine-checked lemmas are Lean 4/Mathlib, machine-checked modulo classical Bessel inputs carried as named hypotheses.

1 Introduction

This paper consolidates the resolution status of a two-part positivity problem that arises as the last analytic gap in a surface (character) expansion for two-dimensional $SU(2)$ lattice gauge theory [5, 6, 7]. The problem is stated entirely in terms of modified Bessel functions and trigonometric series, and we treat it as such; no claim of physical consequence beyond the stated expansion is made, and importance is not inherited by thematic proximity.

Throughout, $\beta > 0$, $I_m = I_m(\beta)$, and

$$a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2], \quad b_m = mI_m^4, \quad m \geq 1. \quad (1)$$

Both sequences decay super-exponentially, so

$$F_A(t) = \sum_{m \geq 1} a_m \sin(mt), \quad F_B(t) = \sum_{m \geq 1} b_m \sin(mt), \quad E(t) = \frac{F_A(t)}{2F_B(t)}, \quad (2)$$

are real-analytic on $\mathbb{R}$, and the Wronskian $W = 2(F'_A F_B - F_A F'_B)$ satisfies $W = 4F_B^2(E)'$ wherever $F_B \neq 0$ [7].

Conjecture 1 (global ratio monotonicity). *For every $\beta > 0$: (i) $F_B > 0$ on $(0, \pi)$; (ii) $E' < 0$ on $(0, \pi)$.*

Part (i) is proved below (twice). Part (ii) is proved for $0 < \beta \leq 3$ (Theorem 18; exact skeleton plus the archived certificate `certify_thmB.py` and its Arb twin), is supported by an exact structural reduction and a verified truth map for $\beta \geq 15$, and is otherwise open; Section 9 states the precise two-scale closure map. The conjecture was born inside this programme (it is the global form of an ε -window positivity required by the surface expansion); no external pedigree is claimed.

Convention 2 (tricotomy). Every assertion below carries one of three labels. *Exact*: a proved identity or inequality. *Certified*: a numerical statement rigorously enclosed by interval arithmetic with outward rounding, together with complete analytic bounds for domains and truncations. Precision-stability re-runs and independent implementations are *software audits*, recorded separately with their declared coverage: the certified negatives `certify_time3_negative` and `certify_bridge_matrix` carry a *full twin audit* (nested re-computation plus a complete independent Arb implementation, both transcripts archived); for `certify_thmB` both implementations exist and the Arb transcript is archived — its row is *amber* only because the `mpmath.iv` canonical transcript is still in progress; the $[6, 15]$ coverage will carry a *planned pre-registered sampled independent audit* (Remark 12). *Verified*: high-precision numerics with stated grids and precision policies, not a proof. No verified statement is used as a hypothesis of an exact one.

2 The theorem and the relay

Theorem 3 (the Surface Theorem; [SLOT: seal completion]). *For every $\beta > 0$: (i) $F_B > 0$ on $(0, \pi)$, and (ii) $E' < 0$ on $(0, \pi)$.*

Part (i) is Theorem 15. Part (ii) is proved by the following relay. Rows marked [SLOT] are pending; rows marked *amber* have one archived witness with the second in progress; all other rows are complete with the stated witnesses.

range	method	artifact	transcript	state
$(0, \frac{1}{20}]$	small- β lemma	Lem. 19	—	complete
$[\frac{1}{20}, 3]$, all t	pair skeleton	<code>certify_thmB</code>	Arb	<i>amber</i>
$[3, 6]$, $t \in [0.6, \pi - \frac{1.5}{\beta}]$	series intervals	<code>certify_bulk</code>	Arb	<i>amber</i>
$[6, 15]$, $t \in [0.6, \pi - \frac{1.5}{\beta}]$	exp-factored integrator	<code>exp_integrator</code>	[SLOT]	local box: Prop. 11
$[15, \infty)$, bulk	analytic relay	[SLOT]	—	pending
$t \in (0, 0.6]$, $\beta \geq 3$	sinc + minor criterion	[SLOT: sinc-cert]	—	design ratified
$t \in [\pi - \frac{C}{\beta}, \pi)$	c_3 + radius	[SLOT: 4 const.]	—	A verified

Splice boundaries are explicit and overlapping: $\beta = 3, 6, 15$; $t = 0.6$ and $t = \pi - C/\beta$, with $C_{\text{WIN}} = 1.5$ a candidate pending $C_{\text{mirror}} \leq 1.5 \leq C_{\text{quad}}$. The remainder of this manuscript is the completed material, verbatim from the consolidation paper.

Remark 4 (supersession of the e_4 route). An earlier plan closed the block $t \in (0, 0.6]$, $\beta \geq 3$ by an e_4 -coefficient lemma with a fixed radius; it is superseded, not silently deleted. A feasibility check

showed the sinc criterion below dominating with ratios from 15 up to 10^{80} across $\beta \in [3, 15]$ for $t \leq 0.3$ (*verified*, design-grade), while the apparent infeasibility of $(0, 0.6]$ belonged to the cancellation of the series tool, which the exp-factored integrator does not suffer. The ratified route has three components: the exact sinc positivity (Lemma 5), cheap one-dimensional-in- β minor-criterion certificates on $(0, t_*(\beta)]$, and the extension of the integrator strip from $t = 0.6$ down to $t_*(\beta) \approx 0.45$ with splice overlap. The slot's pending content is the full minor criterion with its uniform overlap; the e_4 lemma is retired.

Lemma 5 (exact; sinc positivity). *Let $1 \leq m < n$ and $T_{mn}(t) = (m - n) \sin((m + n)t) + (m + n) \sin((n - m)t)$ as in Section 4. Then $T_{mn}(t) > 0$ for $0 < t \leq \pi/(m + n)$.*

Proof. Write $s = m + n$, $p = n - m \geq 1$, so $T_{mn}(t) = s \sin(pt) - p \sin(st)$. Since $\sin(x)/x$ is strictly decreasing on $(0, \pi]$ and $0 < pt < st \leq \pi$, we get $\sin(pt)/(pt) > \sin(st)/(st)$, i.e. $st \sin(pt) > pt \sin(st)$; dividing by $t > 0$ gives $s \sin(pt) > p \sin(st)$. $\square$

Lemma 6 (relay criterion for $\beta \in [6, 15]$). *Let $\langle f \rangle = \iint_{[-\pi, \pi]^2} K f ds d\alpha$ with $K = I_1(2\beta R)/R > 0$, and let $\bar{E}$ be any rational constant. Then $E' < 0$ at (t, β) if and only if $\langle D \rangle > 0$ and*

$$W_c := \langle N_t \rangle \langle D \rangle + \langle (N - \bar{E}D) \partial_t \log K \rangle \langle D \rangle - (\langle N \rangle - \bar{E} \langle D \rangle) \langle D \partial_t \log K \rangle < 0$$

(the $\bar{E}$ -terms cancel identically). *The certificate encloses the five integrals by exp-factored interval cells: $z = 2\beta R$ is linearized at the exact rational cell center with interval-gradient remainder; the linear exponential integrates exactly; and the two scaled Bessel factors are $A(z) = e^{-z} I_1(z)/z$ — strictly decreasing, since $(\log A)' = I_0/I_1 - 1 - 2/z < 0$ follows from the classical lower bound at $\nu = 0$ (Amos [1]; [2], Thm. 1) and $\sqrt{1+z^2} \leq z + 1$ — enclosed by pure endpoint evaluation, and $B(z) = e^{-z} H(z)$, enclosed unconditionally by $[e^{-z_{\text{lo}}} H(z_{\text{lo}}), e^{-z_{\text{hi}}} H(z_{\text{hi}})]$; the identity $K \partial_t \log K = 32\beta^3 \cos(t/4) (\cos(t/4))' (1 - P - Q) H(z)$ removes $I_1(z)/z$ from the covariance integrands exactly. The (t, β) -box layer is Proposition 11; the global $[6, 15]$ coverage remains the pending content of the relay row.*

Lemma 7 (exact; the S -free relative criterion). *Let $S = \sin(t/2)$, $E = \langle N \rangle / \langle D \rangle$, and suppose $\langle D \rangle > 0$. Then for all $t \in (0, \pi)$ and $\beta > 0$,*

$$\frac{E'(t, \beta)}{-\frac{1}{2}S} = \frac{\langle \cos 2s + \cos \alpha \cos s \rangle + 8\beta^3 \iint (N - ED)(1 - P - Q) H(z) ds d\alpha}{\langle D \rangle},$$

and consequently

$$E' < -\frac{1}{4}S \iff \mathcal{C} := \langle \cos 2s + \cos \alpha \cos s \rangle + 8\beta^3 \iint (N - ED)(1 - P - Q) H - \frac{1}{2} \langle D \rangle > 0.$$

Proof. Since $D_t = 0$, $E' = [\langle N_t \rangle + \iint (N - ED) \partial_t K] / \langle D \rangle$. The first term factors S exactly: $N_t = -\frac{S}{2}(\cos 2s + \cos \alpha \cos s)$. So does the second: $\partial_t K = K \partial_t \log K = 32\beta^3 c c' (1 - P - Q) H(z)$ with $c = \cos(t/4)$, and $c c' = \cos(t/4) \cdot (-\frac{1}{4} \sin(t/4)) = -\frac{S}{8}$ by the double angle, so $\partial_t K = -4\beta^3 S (1 - P - Q) H$. Dividing by $-\frac{1}{2}S$ cancels S identically and yields the display. For the equivalence: the denominator $-\frac{1}{2}S$ is negative on $(0, \pi)$, so $E' < -\frac{1}{4}S$ is the lower bound $E'/(-\frac{1}{2}S) > \frac{1}{2}$, i.e. $\mathcal{C} > 0$. $\square$

Corollary 8 (exact; the bilinear quotient-free form). *With $\Phi := \cos 2s + \cos \alpha \cos s$ (the letter A stays reserved for the scaled Bessel factor $e^{-z} I_1(z)/z$), $J_N := \iint N(1 - P - Q) H ds d\alpha$, $J_D := \iint D(1 - P - Q) H ds d\alpha$, and $\langle D \rangle > 0$,*

$$E' < -\frac{1}{4} \sin(t/2) \iff \hat{\mathcal{C}} := \langle D \rangle \langle \Phi \rangle + 8\beta^3 (\langle D \rangle J_N - \langle N \rangle J_D) - \frac{1}{2} \langle D \rangle^2 > 0.$$

Proof. Multiply $\mathcal{C}$ of Lemma 7 by $\langle D \rangle > 0$ and substitute $E = \langle N \rangle / \langle D \rangle$: the quotient disappears by bilinearity, and the inequality direction is preserved. $\hat{\mathcal{C}}$ is a bilinear combination of five integrals with no quotient anywhere; it is the assembly's sole target. $\square$

Remark 9 (why $\hat{\mathcal{C}}$ is the assembly's object). In $\hat{\mathcal{C}}$ the variable t enters only through $c = \cos(t/4) \in (\frac{\sqrt{2}}{2}, 1)$: no kinematically forced vanishing factor and no singularity anywhere in the relay range — the common linear zero of margin and correction at $t \rightarrow 0$ has been cancelled identically. The endpoint degeneracy is thus removed structurally; uniform positivity still requires explicit uniform estimates, which are the assembly's content. The first-order calibration constant is $a(t) = [\frac{1}{2} + \frac{1}{8c^2}]/c \leq \frac{3\sqrt{2}}{4} \approx 1.061$ (supremum at $c = \sqrt{2}/2$), so the first-order threshold is $2\sup a \approx 2.12$; the target of the analytic relay is $\hat{\mathcal{C}} > 0$ uniformly for $\beta \geq \beta_0$ with the remainder included. At the calibration point $(t, \beta) = (1.5, 8)$ the measured value is $\mathcal{C}/\langle D \rangle \approx 0.417$ (*verified*), against the threshold 0. The built-in unit test: the $1/\beta$ expansion of $\mathcal{C}/\langle D \rangle + \frac{1}{2}$ must reproduce $1 - a(t)/\beta$; the audited quantity is $\beta \cdot (\text{deviation from first order})$, which must converge (an $O(\beta^{-2})$ remainder halves the β -scaled deviation when β doubles, and divides the unscaled deviation by four).

Lemma 10 (exact; far-region pointwise and kernel bounds). (a) $\min_{s,\alpha} \Phi = -\frac{9}{8}$ *exactly*: with $u = |\cos s|$ the minimum of $2u^2 - u - 1$ on $[0, 1]$ is attained at $u = \frac{1}{4}$ and equals $-\frac{9}{8}$. (b) $|D| \leq 2$ and, by Corollary 22 ($|E| < 1$), $|N - ED| \leq |N| + |D| \leq (2|C| + 1) + 2 \leq 5$ and $|1 - P - Q| \leq 1$. (c) On the deficit region $\{(s, \alpha) : 2c - R \geq \Delta_0\}$ one has $z = 2\beta R \leq z_s - 2\beta\Delta_0$ with $z_s = 4\beta c$, and since $A(z) = e^{-z}I_1(z)/z$ is strictly decreasing with $A(0^+) = \frac{1}{2}$,

$$K = 2\beta e^z A(z) \leq \beta e^{z_s} e^{-2\beta\Delta_0} \quad \text{on the region.}$$

Hence the far region's contribution to each of the five integrals of Corollary 8 is bounded by (pointwise constant) $\times 4\pi^2 \times \beta e^{z_s} e^{-2\beta\Delta_0}$, exponentially small relative to the saddle mass; the explicit comparison constant against $\langle D \rangle$ is part of the assembly (Regions I–II).

Proof. (a) $\cos \alpha \cos s \geq -|\cos s|$ with equality attainable, and $\cos 2s = 2u^2 - 1$; minimizing $g(u) = 2u^2 - u - 1$ gives $g(\frac{1}{4}) = -\frac{9}{8}$. (b) Triangle inequalities; $|N| \leq |C| |\cos 2s| + |\cos \alpha| (|C \cos s| + \sin^2 s) \leq 2|C| + 1$. (c) $R \leq 2c - \Delta_0$ gives $z \leq z_s - 2\beta\Delta_0$; write $K = I_1(z)/R = 2\beta I_1(z)/z = 2\beta e^z A(z)$ and use the monotonicity of A (Section on the two scaled functions) with $A(z) \leq A(0^+) = \frac{1}{2}$ uniformly. $\square$

Proposition 11 (certified; local box: $(t, \beta) \in [1.5, 1.51] \times [8, 8.05]$). For every $(t, \beta) \in [1.5, 1.51] \times [8, 8.05]$ one has $\langle D \rangle > 0$ and $W_c < 0$; hence, by Lemma 6, $E'(t, \beta) < 0$ on this box.

Certificate. Ball-arithmetic machine certificate (python-flint 0.9.0 / Arb). The executed script is identified by SHA-256 prefix **1d888e99** and is archived byte-exact in the repository, together with the self-contained transcript (both file names carry the hash; the transcript's provenance header records the full hash, library versions, and stage parameters). Three stages, one transcript. *Point* (1.5, 8): 1,951,141 cells, $W_c/\langle D \rangle^2 \subset [-0.4546, -0.1752]$. *Stability* (precision 120, $\Delta z \leq 0.12$): 2,500,336 cells, ratio $\subset [-0.4304, -0.1978]$, nested inside the point enclosure. *Box*: union of nine parametric sub-boxes ($\Delta\beta \leq 0.02$, $\Delta t \leq 0.005$), each with strictly negative W_c enclosure and $\langle D \rangle > 0$ (worst W_c upper end $\leq -2.8 \times 10^{21}$), 22,986,387 cells in total, 7558 s. All comparisons are strict ball comparisons; the displayed intervals are outward-rounded supersets of the printed enclosures. This proposition validates the mechanism and calibrates the coverage design; it does *not* close the $[6, 15]$ row, which stays open until the full region $6 \leq \beta \leq 15$, $0.6 \leq t \leq \pi - 1.5/\beta$ is covered. $\square$

Remark 12 (what certifies, and what audits). The mathematical certificate for each sub-box is its *single* rigorous Arb enclosure: outward-rounded ball arithmetic makes one honest enclosure a proof of the strict inequalities, per sub-box. The precision-120 stability stage (with its nesting inside the point enclosure) and the *planned* pre-registered sampled `mpmath.iv` reproduction are *implementation audits*: they test the software that produced the enclosures, not the inequalities themselves, and they are declared as such in the relay table. This fixes the convention for the [6, 15] coverage: one certificate-grade Arb enclosure per sub-box; stability and sampling audit the pipeline.

Lemma 13 (safe square roots in the archived run). *In the archived script (SHA-256 1d888e99), the square-root guard clamps a negative lower endpoint to 0 and, when the upper endpoint ball straddles 0, returns the dyadic upper bound $2^{8-p/2}$ at working precision $p \geq 90$. This bound is valid throughout the run. Indeed, the guarded inputs are true squares: $R^2 = 4[c_0^2(1-P-Q)+PQ]$ with $c_0^2 \leq 1$, $P, Q \in [0, 1]$, and $z^2 = 4\beta^2 R^2$ with $\beta \leq 15$, so $|R^2| \leq 16$, $|z^2| \leq 14400$; the corresponding balls are built by at most twelve correctly rounded operations on factors of unit size, hence $|R_{\text{ball}}^2| < 16.01$ and $|z_{\text{ball}}^2| < 14401 < 2^{14}$ (magnitude means $|\text{mid}| + \text{rad}$). The upper-endpoint ball $u = \text{mid} + \text{rad}$ involves two exact arf-to-ball conversions and one correctly rounded addition, so $\text{rad}(u) \leq \text{ulp}(2^{14}) = 2^{15-p}$. If u straddles 0 then $|\text{mid}(u)| \leq \text{rad}(u)$, hence the true upper endpoint is at most $2\text{rad}(u) \leq 2^{16-p}$ and its square root is at most $2^{8-p/2}$ — exactly the coded constant. (In the definitive script the branch is replaced by `abs_upper()`, an automatic outer bound with no ad-hoc constant; the present lemma covers the archived run.)*

The two scaled functions

The certificate’s precision rests on three one-line facts. (a) $A(z) = e^{-z}I_1(z)/z$ is strictly decreasing on $(0, \infty)$: by the classical lower bound $I_{\nu+1}(z)/I_\nu(z) \geq z/(\nu+1+\sqrt{(\nu+1)^2+z^2})$, valid for $z > 0$ and $\nu \geq 0$ (Amos [1]; both members of the classical two-sided pair are proved in [2], Theorem 1), taken at $\nu = 0$, together with $\sqrt{1+z^2} \leq z+1$ (square both sides), $I_1/I_0 > z/(z+2)$, which is exactly $(\log A)' = I_0/I_1 - 1 - 2/z < 0$; hence A is enclosed over any z -interval by its two endpoint values. (b) The kernel–score product collapses algebraically, $K \partial_t \log K = 32\beta^3 \cos(t/4)(\cos(t/4))'(1-P-Q)H(z)$ (the factor $I_1(z)/z$ cancels between K and the denominator of the regular score), so the covariance integrands involve only $B(z) = e^{-z}H(z)$. (c) B is enclosed *unconditionally* by $[e^{-z_{\text{hi}}}H(z_{\text{lo}}), e^{-z_{\text{lo}}}H(z_{\text{hi}})]$, since e^{-z} decreases and H (a positive-coefficient entire series) increases; after centering, the B -terms are subdominant and this crude-but-safe bound suffices. Together with the exact linear-exponential cell integrals, every source of exponential interval growth is either integrated analytically or carried by a monotone one-dimensional profile — the certificate is structure, not brute force.

3 The spectral framework

Convention 14 (normalizations). On $(0, \pi)$ define the kernel, entry law, and smoothing operator

$$q_1(u, v) = 2e^{\beta \cos u \cos v} \sinh(\beta \sin u \sin v) = 4 \sum_{m \geq 1} I_m \sin(mu) \sin(mv), \quad (3)$$

$$s_1(u) = \frac{\beta}{2} \sin u e^{\beta \cos u} = \sum_{m \geq 1} m I_m \sin(mu), \quad (4)$$

$$(Qf)(v) = \frac{1}{2\pi} \int_0^\pi q_1(v, u) f(u) du, \quad (5)$$

so that $Q \sin(m \cdot) = I_m \sin(m \cdot)$. Both closed forms are classical consequences of the generating function $e^{\beta \cos \theta} = I_0 + 2 \sum_{m \geq 1} I_m \cos(m\theta)$; (4) follows by differentiation. All ten verification artifacts in the repository use exactly these normalizations.

Since $q_1 > 0$ on $(0, \pi)^2$ and $s_1 > 0$ on $(0, \pi)$, the walk with step kernel q_1 killed on $\{0, \pi\}$ is well defined; $H(\psi) = \beta \sin(\psi/2) I_1(2\beta \cos(\psi/2))$ and $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$ appear below. Note $F_B = Q^3 s_1$ and, by the sine orthogonality computation of Lemma 21, F_A is the companion cosine-weighted integral.

4 Theorem A: positivity of F_B

Theorem 15 (exact). $F_B(t) > 0$ for all $t \in (0, \pi)$ and all $\beta > 0$.

First proof (kernel positivity). $F_B = Q^3 s_1$ by (5) and $b_m = m I_m^4$. Each application of Q integrates a positive function against the strictly positive kernel (3); $s_1 > 0$ on $(0, \pi)$. Hence $F_B > 0$ pointwise. $\square$

Second proof (symmetric-decreasing convolution). With $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$ and $P_4 = K_2 * K_2$ (circle convolution; Section 7), $F_B = -\frac{1}{2} P_4'$. The function K_2 is even and *strictly decreasing in $|\varphi|$ on $[0, \pi]$* (I_0 strictly increasing, $\cos(\varphi/2)$ strictly decreasing). By the layer-cake representation, $K_2 = \int_0^\infty \mathbf{1}_{A_\lambda} d\lambda$ with A_λ centered arcs whose half-lengths $a(\lambda)$ fill $(0, \pi)$ continuously (strict decrease). For centered arcs, $(\mathbf{1}_A * \mathbf{1}_B)(t) = |A \cap (t + B)|$ is even and nonincreasing in $|t|$ on $[0, \pi]$, and *strictly decreasing on $||a - b|, \min(a + b, 2\pi - a - b)|$* . Hence $P_4(t) = \iint |A_\lambda \cap (t + A_\mu)| d\lambda d\mu$ is even, nonincreasing on $[0, \pi]$, and strictly decreasing on $(0, \pi)$ because for every $t \in (0, \pi)$ a positive measure of pairs (λ, μ) has $|a(\lambda) - a(\mu)| < t < \min(a + b, 2\pi - a - b)$ (the half-lengths fill $(0, \pi)$). At the derivative level: each overlap $t \mapsto |A_\lambda \cap (t + A_\mu)|$ is Lipschitz with a.e. derivative -1 in the strict regime and 0 otherwise (on $[0, \pi]$); since P_4 is analytic, dominated convergence gives $P_4'(t) = \iint \partial_t |A_\lambda \cap (t + A_\mu)| d\lambda d\mu \leq -\mu\{(\lambda, \mu) : \text{strict at } t\} < 0$ pointwise on $(0, \pi)$. Thus $F_B = -\frac{1}{2} P_4' > 0$ on $(0, \pi)$. $\square$

5 Theorem B: ratio monotonicity for $0 < \beta \leq 3$

Lemma 16 (exact; Wronskian identity). *For real arrays with $\sum m(|a_m| + |b_m|) < \infty$,*

$$W := 2(F_A' F_B - F_A F_B') = \sum_{1 \leq m < n} c_{mn} T_{mn}(t), \quad c_{mn} = a_m b_n - a_n b_m, \quad (6)$$

where $T_{mn}(t) = (m - n) \sin((m + n)t) + (m + n) \sin((n - m)t)$.

Proof. Product-to-sum: $T_{mn} = 2[m \cos(mt) \sin(nt) - n \sin(mt) \cos(nt)] =: 2G(m, n)$, antisymmetric with $G(n, n) = 0$; since c_{mn} is antisymmetric, $\sum_{m < n} c_{mn} 2G = 2 \sum_{m, n} a_m b_n G(m, n) = 2[F_A' F_B - F_A F_B']$ (first recorded in [7]). $\square$

The coefficient input is the weighted Turán-type lemma (Appendix A, self-contained): $\varphi_m = a_m/b_m$ is strictly increasing in $m \geq 1$ for every $\beta > 0$, hence *every minor is negative*: $c_{mn} < 0$ for $m < n$ (exact).

Lemma 17 (exact; the pair identities and bounds). *For $t \in (0, \pi)$, with $p = n - m$, $q = n + m$:*

(a) $T_{12} = 4 \sin^3 t$; $T_{13} = 16 \sin^3 t \cos t$; $T_{24} = 8 \sin^3(2t)$, so $|T_{13}| \leq 16 \sin^3 t$ and $|T_{24}| \leq 64 \sin^3 t$.

(b) $T_{m,m+1} \geq 0$ for every $m \geq 1$.

(c) $|T_{mn}| \leq \frac{\pi^3}{48} pq(q^2 + p^2) \sin^3 t$ for all $m < n$.

Proof. (a) $3 \sin u - \sin 3u = 4 \sin^3 u$ gives T_{12} (at $u = t$) and T_{24} (at $u = 2t$); $T_{13} = 4 \sin 2t - 2 \sin 4t = 4 \sin 2t(1 - \cos 2t) = 16 \sin^3 t \cos t$. (b) $T_{m,m+1} = (2m+1) \sin t - \sin((2m+1)t) \geq 0$ since $|\sin(qt)| \leq q \sin t$ on $(0, \pi)$ (induction on q). (c) T_{mn} and its first two derivatives vanish at $t = 0$ and $t = \pi$ (the parities of $m \pm n$ agree), and $|T_{mn}'''| \leq pq^3 + qp^3 = pq(q^2 + p^2)$; hence $|T_{mn}(t)| \leq \frac{pq(q^2+p^2)}{6} \min(t, \pi-t)^3$, and $\sin t \geq \frac{2}{\pi} \min(t, \pi-t)$ gives the claim. $\square$

Theorem 18 (exact modulo the certified computation `certify_thmB.py`). For $0 < \beta \leq 3$: $W < 0$ on $(0, \pi)$, hence $E' < 0$ on $(0, \pi)$.

Proof. Drop the adjacent pairs $m \geq 2$ (each contributes $c_{m,m+1}T_{m,m+1} \leq 0$ by Lemma 17(b) and $c < 0$); keep T_{12}, T_{13}, T_{24} exactly and majorize every other pair by Lemma 17(c). Pointwise in t ,

$$\frac{W(t)}{\sin^3 t} \leq \text{Crit}(\beta) := -4|c_{12}| + 16|c_{13}| + 64|c_{24}| + \frac{\pi^3}{48} \sum_{\substack{p \geq 2 \\ (m,n) \neq (1,3), (2,4)}} pq(q^2 + p^2) |c_{mn}|.$$

The certificate `certify_thmB.py` (Section 10) encloses Crit in interval arithmetic over adaptive β -boxes covering $[1/20, 3]$ (I_m increasing in β gives rigorous boxes; cutoff $M = 60$ with the tail itself computed in intervals and closed by a derived, per-box-asserted geometric ratio) and certifies $\text{Crit} < 0$ on all of $[1/20, 3]$; the script then re-certifies *every* box at precision $+70$ before reporting success (the box count is invocation- and implementation-dependent; the independent Arb implementation `certify_thmB_arb.py` certifies the same coverage with 86 boxes and a full stability pass, transcript archived). For $0 < \beta \leq 1/20$, Lemma 19 applies. $\square$

Lemma 19 (exact; small β). For $0 < \beta \leq 1/20$: $\text{Crit}(\beta) < 0$.

Proof. Write $\kappa = \beta/2 \leq 1/40$, $L_m = \kappa^m/m!$, $U_m = e^* L_m$ with $e^* = e^{\beta^2/4} \leq e^{1/1600} < 1.001$; then $L_m \leq I_m \leq U_m$ (positive series). *Leading minor:* $a_2 b_1 \geq (I_1^2 I_2^2)(I_1^4) \geq L_1^6 L_2^2 = \kappa^{10}/4$ and $a_1 b_2 = (2I_1^2 I_2^2)(2I_2^4) \leq 4U_1^2 U_2^6 = (e^*)^8 \kappa^{14}/16$, so

$$|c_{12}| \geq \frac{\kappa^{10}}{4} \left(1 - \frac{(e^*)^8 \kappa^4}{4}\right) \geq \frac{\kappa^{10}}{4} (1 - 10^{-6}).$$

The three visible positive terms: $a_3 \leq I_3^2 \cdot (2I_2^2 + 4I_4^2) \leq (e^*)^4 \kappa^{10}/24 \cdot (1 + 10^{-4})$ and $b_1 \leq (e^*)^4 \kappa^4$ give $16|c_{13}| \leq 16(a_3 b_1 + a_1 b_3) \leq 0.70 \kappa^{14}$; $64|c_{24}| \leq 64(a_4 b_2 + a_2 b_4) \leq 10^{-3} \kappa^{14}$ (both products have κ -degree ≥ 22); the largest remaining weighted pair is $(1, 4)$: $\frac{\pi^3}{48} \cdot 510 \cdot (a_4 b_1 + a_1 b_4) \leq 0.2 \kappa^{18} \leq 10^{-6} \kappa^{14}$. *All other pairs* (those with $s := m + n \geq 6$, $p \geq 2$): using $a_n b_m + a_m b_n \leq 2mn(e^*)^8 \kappa^{4s-2} [((m-1)!m!)^{-2} n!^{-4} + ((n-1)!n!)^{-2} m!^{-4}] \leq s^2(e^*)^8 \kappa^{4s-2}$ (the factorial bracket is ≤ 2 and $2mn \leq s^2/2$), $pq(q^2 + p^2) \leq 2s^4$, and at most s pairs per level,

$$\frac{\pi^3}{48} \sum_{s \geq 6} (\text{level } s) \leq \frac{\pi^3}{48} (e^*)^8 \sum_{s \geq 6} 2s^7 \kappa^{4s-2} \leq 0.66 \kappa^{22} \cdot 6^7 \sum_{j \geq 0} [(7/6)^7 \kappa^4]^j \leq 3.7 \times 10^5 \kappa^{22} \leq 10^{-7} \kappa^{14},$$

where the level ratio $s \mapsto s + 1$ is at most $(7/6)^7 \kappa^4 < 10^{-5} < \frac{1}{2}$ (so the geometric closure is valid), and $\kappa^8 \leq 40^{-8}$ absorbs the constant. Collecting:

$$\text{Crit} \leq -\kappa^{10}(1 - 10^{-6}) + 0.71 \kappa^{14} = -\kappa^{10}(1 - 10^{-6} - 0.71 \kappa^4) < 0,$$

since $\kappa^4 \leq 40^{-4} < 4 \cdot 10^{-7}$. (All numbered constants above are crude upper counts; each step loses factors, none gains.) $\square$

Remark 20 (verified; the sharp family and saturation). The sharper per-pair bound $|T_{mn}| \leq \frac{pq(q^2-p^2)}{6} \sin^3 t$ and the adjacent floor $T_{m,m+1} \geq 2m \sin^3 t$ (note the sign: the floor is a *lower* bound on a positive quantity) are verified far beyond the range used ($q \leq 151$; 13 pairs to (10, 30), max ratio 1.0) and would extend Theorem 18 to $\beta \approx 3.5$; their per-pair certification is a scheduled upgrade. Beyond that the decomposition is *saturated*: with the exact far block (cancellation included) the criterion fails at $\beta = 6$ by $+2.2 \times 10^{14}$ while the true $W/\sin^3 t$ remains negative (-6.9×10^{12}) — a verified adversarial finding about the method, and the reason the large- β closure uses different machinery (Sections 7–9).

6 The bridge structure of E (exact)

Let $k = q_1 \circ q_1$ denote the two-step kernel, $k(\psi, t) = 4 \sum_{m \geq 1} I_m^2 \sin(m\psi) \sin(mt)$, and define

$$\omega_t(d\psi) \propto (Qs_1)(\psi) k(\psi, t) d\psi \quad \text{on } (0, \pi), \quad (7)$$

the law of the time-2 position (midpoint) of the four-step bridge entering from 0 with law s_1 and conditioned to end at t .

Lemma 21 (exact). $\int_0^\pi (Qs_1) k(\cdot, t) d\psi = 2\pi F_B(t)$ and $\int_0^\pi \cos \psi (Qs_1) k(\cdot, t) d\psi = \pi F_A(t)$. Hence, as an identity,

$$E(t) = \mathbb{E}_{\omega_t}[\cos \psi] = \frac{F_A(t)}{2F_B(t)}.$$

Proof. $(Qs_1)(\psi) = \sum m I_m^2 \sin(m\psi)$. By sine orthogonality the first integral picks the diagonal, giving $2\pi \sum m I_m^4 \sin(mt)$. In the second, $\cos \psi \sin(m\psi) \sin(n\psi)$ integrates to $\frac{\pi}{4}(\delta_{n,m+1} + \delta_{n,m-1})$, which produces exactly the weights $(n-1)I_{n-1}^2 + (n+1)I_{n+1}^2$ of a_n . $\square$

Corollary 22 (exact; $|E| < 1$). For all $t \in (0, \pi)$ and $\beta > 0$, $|E(t, \beta)| < 1$. Indeed $q_1(u, v) = 2e^{\beta \cos u \cos v} \sinh(\beta \sin u \sin v) > 0$ on $(0, \pi)^2$ and $s_1 > 0$, so $(Qs_1) > 0$ and $k = q_1 \circ q_1 > 0$ on $(0, \pi)$; hence ω_t in (7) is a genuine probability measure and $E = \mathbb{E}_{\omega_t}[\cos \psi] \in (-1, 1)$.

Lemma 23 (exact; entry law). $\partial_t k(0^+, \psi) = 2H(\psi)$, where $H(\psi) = \beta \sin(\psi/2) I_1(2\beta \cos(\psi/2))$.

Proof. Differentiate in ψ the squared Graf identity $I_0^2 + 2 \sum_{m \geq 1} I_m^2 \cos(m\psi) = I_0(2\beta \cos(\psi/2))$ [4, §10.23(ii)] and compare with $\partial_t k(0^+, \psi) = 4 \sum m I_m^2 \sin(m\psi)$. $\square$

Proposition 24 (certified negative; no monotone full-path coupling). At $\beta = 20$ the time-3 marginal $\rho_t(u) \propto (Q^2 s_1)(u) q_1(u, t)$ is not stochastically monotone in t : with $a = \frac{4802}{10^4}$, $t_1 = \frac{29621}{10^4}$, $t_2 = \frac{30070}{10^4}$, the tail masses satisfy $T(t_2, a) < T(t_1, a)$, with certified enclosure

$$T(t_2, a) - T(t_1, a) \in [-2.5299150980081786690028107, -2.5299150980081786690028074] \times 10^{-10}.$$

Consequently no monotone coupling of full bridge paths exists at $\beta = 20$. The certificate uses exact termwise tails ($J(m, n, a) = \int_a^\pi \sin m u \sin n u du$ in closed form: no quadrature error), positive-series Bessel enclosures with geometric tails, explicit truncation bounds, nested passes $(M, \text{prec}) = (120, 350) \subset (140, 500)$, and two independent interval implementations (*mpmath.iv*; *Arb* via *python-flint*).

Proposition 25 (certified negatives; the weight–kernel matrix). *At $\beta = 20$, for every entry weight $Q^{k-1}s_1$ ($k = 1, 2, 3, 4$) the marginal one q_1 -step from the moving endpoint fails stochastic monotonicity, with certified witnesses sharing $(t_1, t_2) = (3, 61/20)$: $k = 1$ (two-step bridge midpoint, the minimal counterexample), $a = 9/10$, margin $\approx -1.34 \times 10^{-2}$; $k = 2$, $a = 3/4$, $\approx -1.80 \times 10^{-6}$; $k = 4$, $a = 3/10$, $\approx -4.75 \times 10^{-13}$ ($k = 3$ is Proposition 24). Two smoothing steps repair all tested weights (verified); one never does. The statement is relative to the natural weight family: k_2 possesses its own total-positivity corner ($\beta \gtrsim 8$) reachable by two-atom adversarial weights.*

Lemma 26 (exact; the 3/2 mechanism). $\lim_{u \downarrow 0} \frac{\partial_u \partial_t \log q_1(u, t)}{\beta \sin t u} = 1 + \frac{2\beta}{3} \cos t$. For $t > \pi/2$ the corner sign change of $\partial_u \partial_t \log q_1$ occurs exactly when $\beta |\cos t| > 3/2$ — the same threshold as the TP_2 failure of q_1 .

Proof. $\partial_t \log q_1 = -\beta \sin t \cos u + \beta \cos t \sin u \coth(\beta \sin t \sin u)$ (direct differentiation of (3)); expand $\coth z = z^{-1} + z/3 + O(z^3)$ and differentiate in u ; the stated limit is a one-line computation (symbolic residual zero). $\square$

Remark 27 (verified; the mechanism reading). The identification of this local defect as *the* engine of Propositions 24–25 — sine-type entry weights, vanishing linearly at 0 with mass on both sides of the non-monotone region, flip the covariance $\text{Cov}_{\rho_t}(\mathbf{1}_{u \geq a}, \partial_t \log q_1)$, while the flat weight does not — is supported by the full weight–kernel matrix and the β -scan of the violations (onset at $\beta \gtrsim 12$, consistent with the threshold), but is a verified reading, not a theorem.

7 The single-Bessel reduction and the master formula

Lemma 28 (exact; Neumann product formulas). *For $x, y \geq 0$, with $w = \sqrt{x^2 + y^2 + 2xy \cos \alpha}$,*

$$I_0(x)I_0(y) = \frac{1}{2\pi} \int_0^{2\pi} I_0(w) d\alpha, \quad I_0(x)I_1(y) = \frac{1}{2\pi} \int_0^{2\pi} I_1(w) \frac{y + x \cos \alpha}{w} d\alpha.$$

Proof. The first is the integrated Graf/Neumann addition theorem [4, §10.23(ii)]; the second is its ∂_y -derivative. $\square$

Remark 29 (signs). Both formulas extend to all real x, y , as used below (on $[-\pi, \pi]^2$ one has $c_1 = \cos(\varphi/2) \geq 0$ but c_2 may be negative): $w(x, -y, \alpha) = w(x, y, \pi - \alpha)$, and the substitution $\alpha \mapsto \pi - \alpha$ over a full period maps the first identity to itself (both sides even in y) and the second to its negative on both sides simultaneously (I_1 odd; $(-y + x \cos(\pi - \alpha)) = -(y + x \cos \alpha)$).

With $K_2(\varphi) = I_0(2\beta \cos(\varphi/2)) = \sum_{m \in \mathbb{Z}} I_m^2 e^{im\varphi}$ and $P_4 = K_2 * K_2$ (circle convolution), $P_4 = \sum_m I_m^4 e^{imt}$ and $F_B = -\frac{1}{2}P_4'$. Applying Lemma 28 to both convolution factors:

Lemma 30 (exact; master representation). *Let $s = \varphi - t/2$, $c_1 = \cos(\varphi/2)$, $c_2 = \cos((t - \varphi)/2)$, $P = \sin^2(s/2)$, $Q = \sin^2(\alpha/2)$, and*

$$R^2 = c_1^2 + c_2^2 + 2c_1c_2 \cos \alpha = 4[\cos^2(t/4)(1 - P - Q) + PQ]. \quad (8)$$

Then

$$F_B(t) = \frac{\beta \sin(t/2)}{8\pi^2} \iint_{[-\pi, \pi]^2} \frac{I_1(2\beta R)}{R} [\cos^2(s/2) - \sin^2(\alpha/2)] ds d\alpha, \quad (9)$$

and, under the common positive kernel $K = I_1(2\beta R)/R$,

$$E(t) = \frac{\langle \cos(t/2) \cos 2s + \cos \alpha [\cos(t/2) \cos s - \sin^2 s] \rangle}{\langle \cos s + \cos \alpha \rangle}, \quad (10)$$

where $\langle \cdot \rangle$ denotes the $K ds d\alpha$ -integral over $[-\pi, \pi]^2$. The prefactor collapse behind (9) is the sum-to-product symmetrization $\frac{1}{2}[\sin(\frac{t-\varphi}{2})(c_2 + c_1 \cos \alpha) + (\varphi \leftrightarrow t - \varphi)] = \frac{1}{2} \sin(t/2)(\cos s + \cos \alpha)$, and $\cos s + \cos \alpha = 2[\cos^2(s/2) - \sin^2(\alpha/2)]$.

Proof. (i) The squared Graf identity gives $K_2(\varphi) = \sum_m I_m^2 e^{im\varphi}$; convolution multiplies Fourier coefficients, so $P_4 = \sum_m I_m^4 e^{im\varphi}$ and, termwise (super-exponential decay), $F_B = -\frac{1}{2}P_4'$. (ii) Insert Lemma 28 (first formula) with $x = 2\beta c_1$, $y = 2\beta c_2$ into $P_4(t) = \frac{1}{2\pi} \int K_2(\varphi) K_2(t - \varphi) d\varphi$: a two-dimensional integral of $I_0(2\beta R)$ with R as in (8); the (P, Q) form of R^2 is product-to-sum. (iii) Differentiate under the integral (smooth integrand, compact domain): $\partial_t I_0(2\beta R) = I_1(2\beta R) \cdot 2\beta \partial_t R$ and $2R \partial_t R = 2(c_2 + c_1 \cos \alpha) \partial_t c_2$ with $\partial_t c_2 = -\frac{1}{2} \sin((t - \varphi)/2)$. (iv) Symmetrize over $\varphi \leftrightarrow t - \varphi$ (R invariant): with $\sin(x/2) \cos(x/2) = \frac{1}{2} \sin x$ and the angle-addition formula,

$$\begin{aligned} \frac{1}{2} [\sin(\frac{t-\varphi}{2})(c_2 + c_1 \cos \alpha) + \sin(\frac{\varphi}{2})(c_1 + c_2 \cos \alpha)] &= \frac{1}{4} [\sin(t - \varphi) + \sin \varphi] + \frac{1}{2} \sin(\frac{t}{2}) \cos \alpha \\ &= \frac{1}{2} \sin(\frac{t}{2})(\cos s + \cos \alpha), \end{aligned}$$

using $\sin(t - \varphi) + \sin \varphi = 2 \sin(t/2) \cos s$ and $\sin(\frac{t-\varphi}{2})c_1 + \sin(\frac{\varphi}{2})c_2 = \sin(t/2)$. Collecting constants yields (9). The same substitution in the companion representation of F_A (Lemma 21 with Lemma 28, second formula) and cancellation of the common factor $\beta \sin(t/2)$ yields (10). As an end-to-end check, (9) and (10) match the quartic series with ratio 1.000000000000 at $t \in \{0.4, 1.1, 2.3, 2.9\}$, $\beta = 4$. $\square$

Lemma 31 (exact; geometry of the (P, Q) square, $0 < t < \pi$). *On $[0, 1]^2$: the main saddle is $(0, 0)$ with $R = 2 \cos(t/4)$; the secondary saddle is $(1, 1)$ with $R = 2 \sin(t/4)$ and gap $2(\cos(t/4) - \sin(t/4)) > 0$ degenerating only at $t = \pi$; the zero set $\{R = 0\}$ consists of exactly the two corners $(1, 0)$ and $(0, 1)$, and at both of them both brackets of (9) and (10) vanish, so the singular points of K are switched off (the extension $I_1(z)/z \rightarrow \frac{1}{2}$ makes K continuous with value β). At $t = 0$ the zero set degenerates to the edges $P = 1$, $Q = 1$. The saddle deficit is the exact sum of two squares*

$$4 \cos^2(t/4) - R^2 = 4 \cos^2(t/4) \sin^2(s/2) + 4c_1c_2 \sin^2(\alpha/2), \quad c_1c_2 = \cos^2(t/4) - \sin^2(s/2), \quad (11)$$

with sign region $c_1c_2 \geq 0 \iff |s| \leq \pi - t/2$.

Proof. (8) and (11) are product-to-sum computations (symbolic zeros). $R^2 \geq (|c_1| - |c_2|)^2$ with equality forcing $|c_1| = |c_2|$ (i.e. $s = 0$ or $s = \pm\pi$) and $\cos \alpha = \mp \text{sign}(c_1c_2)$, which lands exactly on $(P, Q) \in \{(0, 1), (1, 0)\}$. Bracket vanishing at both points is direct substitution. The Jacobian of $(s, \alpha) \mapsto (P, Q)$ carries the arcsine weight $ds d\alpha = dP dQ / \sqrt{P(1-P)Q(1-Q)}$ (per quadrant), singular at all four corners; near the main saddle one therefore works in (s, α) , and in (P, Q) only for the global geometry. $\square$

Lemma 32 (exact; saddle cancellations). *Write N, D for numerator and denominator brackets of (10), $C = \cos(t/2)$, $c_0 = \cos(t/4)$. At the saddle $(s, \alpha) = (0, 0)$:*

$$\frac{N}{D}\Big|_{(0,0)} = C, \quad N_{\alpha\alpha} - C D_{\alpha\alpha} = 0, \quad N_{ss} - C D_{ss} = -4C - 2,$$

and for $N_t = -\frac{1}{2} \sin(t/2)[\cos 2s + \cos \alpha \cos s]$:

$$N_{t,\alpha\alpha} - m D_{\alpha\alpha} = 0 \quad (m = -\frac{1}{2} \sin(t/2)), \quad N_{t,ss} - m D_{ss} = 2 \sin(t/2).$$

Also exact: $2C + 1 = 4c_0^2 - 1$ and $C c(t) = 2c_0 - 1/(2c_0)$ for $c(t) = (4c_0^2 - 1)/(2c_0 C)$.

Proposition 33 (verified; the second-order laws). *The frozen-kernel expansion built on Lemma 32 (with $\langle s^2 \rangle = 1/(\beta c_0) + O(\beta^{-2})$ from (11)) predicts*

$$E = C \left(1 - \frac{c(t)}{\beta}\right) + O(\beta^{-2}), \quad E' = -\frac{1}{2} \sin(t/2) + \frac{\sin(t/4)(4c_0^2 + 1)}{8\beta c_0^2} + O(\beta^{-2}). \quad (12)$$

The coefficient identities are exact (Lemma 32); the laws with remainder are verified, not yet certified: measured remainders are $\approx 0.15/\beta^2$ with true convergence at fixed t (e.g. β^2 -scaled residues 0.1776/0.1767/0.1762 at $t = 1, \beta = 30/60/120$), and $\beta(E' + \frac{1}{2} \sin(t/2)) \rightarrow \sin(t/4)(4c_0^2 + 1)/(8c_0^2)$ at $t = 1, 2$. The remainder control is exactly what the closure map owes. Note the $1/\beta$ correction to E' is positive: it opposes the target inequality and must be beaten, not used. The α -cancellations imply every kernel correction multiplies a vanishing fluctuation and enters at $O(\beta^{-2})$ only.

Remark 34 (exact decomposition of E'). With the normalized positive measure $d\mu \propto K ds d\alpha$, $E' = \langle N_t \rangle / \langle D \rangle + \langle (N - ED) \partial_t \log K \rangle / \langle D \rangle$, $D_t = 0$, and the regular form $\partial_t \log K = 4cc'(1 - P - Q)[zI_0(z)/I_1(z) - 2]/R^2$, $z = 2\beta R$, extends continuously by $4\beta^2 cc'(1 - P - Q)$ at $R = 0$. Verified split at $(\beta, t) = (5, 1.3)$: $-0.238212 - 0.024552 = -0.262764 =$ the finite difference exactly. Bounds for E' are taken from this exact quotient — never by differentiating an estimate of E .

8 The π -endpoint package

Lemma 35 (exact; telescoping). *With a_m as in (1):*

$$A_\infty := \sum_{m \geq 1} (-1)^{m+1} m a_m = 0, \quad \sum_{m \geq 1} (-1)^{m+1} m^3 a_m = - \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2.$$

Hence $F_A(\pi - \varepsilon) = c_3 \varepsilon^3 + O(\varepsilon^5)$ with

$$c_3 = \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2. \quad (13)$$

Proof. Split $m^r a_m = m^r(m-1)I_{m-1}^2 I_m^2 + m^r(m+1)I_m^2 I_{m+1}^2$ and shift $m \rightarrow m+1$ in the first sum: for $r = 1$ the two weights cancel ($m(m+1) - m(m+1) = 0$); for $r = 3$ they combine to $m(m+1)[m^2 - (m+1)^2] = -m(m+1)(2m+1)$. Absolute convergence justifies the reindexing; $\sin(m(\pi - \varepsilon)) = (-1)^{m+1} \sin(m\varepsilon)$ expands the sine. $\square$

Lemma 36 (exact; integral form and parity). *Let $g(u) = \frac{1}{2}I_1(2\beta \cos u) = \sum_{m \geq 0} I_m I_{m+1} \cos((2m+1)u)$ (addition theorem). Then*

$$c_3 = \frac{S_3 - S_1}{24}, \quad S_1 = \frac{2}{\pi} \int_0^\pi g\left(\frac{\pi}{2} - u\right) g'(u) du, \quad S_3 = \frac{2}{\pi} \int_0^\pi g'\left(\frac{\pi}{2} - u\right) (-g''(u)) du.$$

Moreover $g(-u) = g(u)$ and $g(\pi - u) = -g(u)$ (parity of I_1), hence $g'(\pi - u) = g'(u)$, $g''(\pi - u) = -g''(u)$, and both integrands are invariant under $u \mapsto \pi - u$: the two halves $[0, \pi/2]$ and $[\pi/2, \pi]$ contribute exactly equally, and one may integrate a half range and double. In particular no cancellation between the two phase maxima $u = \pi/4$ and $u = 3\pi/4$ is possible — exactly, not asymptotically.

Proof. The expansion of g is the Graf addition theorem with unit shift. Quarter turn: $\cos((2m+1)(\frac{\pi}{2} - u)) = (-1)^m \sin((2m+1)u)$, so $g(\frac{\pi}{2} - u) = \sum_m (-1)^m I_m I_{m+1} \sin((2m+1)u)$ while $g'(u) = -\sum_k (2k+1) I_k I_{k+1} \sin((2k+1)u)$ and $g''(u) = -\sum_k (2k+1)^2 I_k I_{k+1} \cos((2k+1)u)$, whence $g'(\frac{\pi}{2} - u) = -\sum_k (-1)^k (2k+1) I_k I_{k+1} \cos((2k+1)u)$. Orthogonality of $\{\sin((2m+1)u)\}$ and $\{\cos((2m+1)u)\}$ on $[0, \pi]$ (norm $\pi/2$) gives $S_1 = -\sum_m (-1)^m (2m+1) (I_m I_{m+1})^2$ and $S_3 = -\sum_m (-1)^m (2m+1)^3 (I_m I_{m+1})^2$, so

$$\begin{aligned} \frac{S_3 - S_1}{24} &= \sum_{m \geq 0} (-1)^{m+1} \frac{(2m+1)^3 - (2m+1)}{24} (I_m I_{m+1})^2 \\ &= \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2 = c_3, \end{aligned}$$

using $(2m+1)^3 - (2m+1) = 4m(m+1)(2m+1)$ and the vanishing of the $m = 0$ term. The parity claims follow from $I_1(-z) = -I_1(z)$ applied to $g(\pi - u) = \frac{1}{2}I_1(-2\beta \cos u)$, and by differentiating the two symmetry relations. $\square$

Proposition 37 (verified; prefactor law). $c_3 > 0$ for $\beta \in \{1, 5, 20, 40, 80, 120\}$ at cancellation-adjusted precision (the alternating sum cancels at scale $e^{-(4-2\sqrt{2})\beta}$; evaluations use $\text{dps} \geq 0.6\beta + 50$), and

$$c_3 \sim A \beta^{3/2} e^{2\sqrt{2}\beta}, \quad A = \frac{1}{24 \cdot 2^{1/4} \pi^{3/2}} = 0.0062922 \dots,$$

with measured ratios 0.0058646, 0.0060049, 0.0060996, 0.0061475 at $\beta = 40, 60, 90, 120$ and $O(1/\beta)$ gaps. The certified large- β branch (pending) must produce the signed prefactor with an explicit B : $I_{\text{half}} \geq \frac{A}{2} \beta^{3/2} e^{2\sqrt{2}\beta} (1 - B/\beta) - T(\beta)$, $c_3 = 2I_{\text{half}}$, read β^* , and close $[15, B_{\text{fin}}]$, $B_{\text{fin}} = \max\{\beta^*, 10C\}$, by intervals.

9 The closure map and the candidate threshold

Proposition 38 (verified truth map; all statements on the stated finite grids). *Let $g(t, \beta) = E' + \frac{1}{4} \sin(t/2)$. At $\beta = 15$, $g < 0$ at every point of a 13-point grid spanning $[0.05, \pi - 0.09]$ (independently reproduced on a much finer mesh at 110 digits), with thinnest margin -0.00573 at $t = 0.05$ (on the seal window $[0.05, \pi - 0.10]$) and -0.00216 at $\pi - 0.09$; the crossing sits at $\pi - t^* = 0.0890502225$. At $\beta = 12$ the truth itself fails at $\pi - 0.10$ ($g = +0.025$); crossings scale as the mirror window: $\pi - t^* = 0.1155, 0.0891, 0.0644$ at $\beta = 12, 15, 20$, matching $\sqrt{2}/\beta$. A sweep $\beta = 15 \dots 60$ keeps the maximum at $t = 0.05$, drifting to -0.0061 : strong uniformity evidence, not proof.*

Conjecture 39 (the pending rows of the relay, quantified). *Part (ii) of Conjecture 1 holds with the two-scale decomposition: (a) $0 < \beta \leq 3$ by Theorem 18; (b) $\beta \in [3, 15]$ by certified evaluation (the holonomic structure of Section 10 makes the compact cheap); (c) $\beta \geq 15$ by three uniform blocks (windows per the LIVE map of Section 2, superseding earlier drafts): left edge $(0, 0.6]$ (by the ratified sinc route of Remark 4: the exact sinc positivity of Lemma 5, the certified minor criterion with control of the high-frequency tail $m+n > \pi/t$, and the uniform splice overlap with the integrator strip extended down to $t_*(\beta) \approx 0.45$); bulk $[0.6, \pi - C/\beta]$ (enclosure of $\langle s^2 \rangle$ and of the covariance of Remark 34, with the five prototype-to-certificate conversions); right edge with moving boundary: assembly to $\pi - C/\beta$, then the quadratic regime of c_3 on $[\pi - C/\beta, \pi]$ with $C_{\text{mirror}} \leq C \leq C_{\text{quad}}$ (a fixed quadratic window is provably impossible: $\kappa \approx 0.11\beta$ grows while $E(\pi - 0.10)$ stays bounded; retention $E/\kappa d^2$ at $d = 2/\beta$ is 0.896, 0.892, 0.890 at $\beta = 20, 40, 80$ — the validity radius scales as $1/\beta$). The named constants still owed: B , β^* , C_{quad} , C_{mirror} , and the remainder function $T(\beta)$.*

The first-order competition is not the obstacle: the two-term law (12) requires only $\beta > (1 + 1/(4c_0^2))/c_0 \leq 3\sqrt{2}/2 \approx 2.121$; everything above that is rigor slack. The candidate $\beta_0 = 15$ is near truth-optimal for its edge window (Proposition 38), not padded.

10 Verification, certificates, and reproducibility

All numerical artifacts live in the repository

<https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>

(directories `scripts/`, `docs/`); the immutable commit hash for the version audited with this manuscript is recorded in the repository’s hash registry (`docs/SURFACE-CLOSURE-NOTES.md`) and in the arXiv comment field on submission. Each certificate embeds its pinned statement and self-verifies on every execution.

Certificates (interval arithmetic, outward rounding, nested or stability passes). `certify_time3_negative.py` and its Arb twin (Proposition 24); `certify_bridge_matrix.py` and twin (Proposition 25); `certify_thmB.py` (Theorem 18: adaptive β -boxes covering $[1/20, 3]$; tails computed in intervals and closed at the level of *complete shells*: each pair maps into the next shell with term ratio r derived from the termwise-exact $I_{n+1}(x) \leq xI_n(x)/(2(n+1))$, and the one new pair per shell is bounded by a double index shift, giving shell ratio $2r$, asserted $< \frac{1}{2}$ per box before the geometric closure; a built-in full stability pass at precision +70; the Arb twin `certify_thmB_arb.py` independently certifies the full coverage — 86 boxes, both passes, transcript archived; the `mpmath.iv` canonical transcript is committed on completion of its run, as recorded in the repository hash registry). Bessel enclosures use the positive series with an exact-integer stopping rule and geometric tails; floats appear only in loop-termination heuristics. Nested enclosures prevent silent numerical or truncation degradation; statement auditing and the independent Arb implementation guard against common-mode errors.

Verified computations. The master-formula and reduction identities (ratios 1.0 to 12 digits); the truth map of Proposition 38 (dps 80, confirmed independently at 110 digits); the c_3 ladder (Proposition 37); the term-split coefficients of (12) at $\beta = 60, 120, 240$.

Precision policies (each traceable to a caught artifact): $\text{dps} \geq 2.2\beta + 40$ for any evaluation with $t \geq 2$, $\beta > 30$ (mirror cancellation); $\text{dps} \geq 0.6\beta + 50$ for c_3 -type alternating sums; series cutoffs scale with the maximal argument, and scaled-asymptotic evaluation of $e^{-z}I_n(z)$ is the

default for $z > 35$; every measured coefficient carries a β -scaling sanity check; whatever can be computed from a positive integral representation is.

Audit trail. Fifteen numerical or formulation artifacts (“ghosts”) were caught during fabrication and writing by the three-way review protocol; Appendix B tabulates them. We consider the table part of the result: it is the reproducibility record that makes the certificates credible.

Machine-checked lemmas. The unit-step and Turán-type coefficient inequalities behind Theorem 18 are formalized in Lean 4/Mathlib (`PhiMonotone.lean`, `FHBesselAmos.lean`, `WronskianIdentity.lean`; axiom oracle `[propext, Classical.choice, Quot.sound]`, no sorry). Mathlib has no Bessel functions: what is machine-checked is the ordered-field implication, with the analytic inputs (three-term recurrence [4, 10.29.1]; the Amos-type bound of [1, 2, 3], cited verbatim, shift $\nu \rightarrow \nu + 1$ as in Appendix A) carried as named hypotheses. Correct claim everywhere: *machine-checked modulo classical Bessel inputs carried as named hypotheses.*

Acknowledgements and provenance

The conjecture, the bridge reading, the reduction, and all certificates were produced inside an audit-first programme in which three independent reviewing voices verify every claim before acceptance; scores, objections, and the complete ghost ledger are archived with the repository history. Nothing in this paper rests on trust in any single computation: every load-bearing negative has two witnesses, and every identity marked exact has a symbolic-zero transcript.

A The weighted Turán-type lemma, self-contained

Theorem 40 (exact). *For every $x > 0$ and every real $m \geq 1$, with $\varphi_m = a_m/b_m$ built from (1) at argument x : $\varphi_m < \varphi_{m+1}$. Equivalently, with $\rho_\mu = I_{\mu+1}/I_\mu$,*

$$\frac{1}{m} \left(\frac{m-1}{\rho_{m-1}^2} + (m+1)\rho_m^2 \right) < \frac{1}{m+1} \left(\frac{m}{\rho_m^2} + (m+2)\rho_{m+1}^2 \right).$$

Proof. Two classical inputs, valid for all real orders: the three-term recurrence $\rho_{m-1}^{-1} = \rho_m + 2m/x$, $\rho_{m+1} = \rho_m^{-1} - 2(m+1)/x$ [4, 10.29.1], and the Amos-type bound

$$\rho_m(x) < U_m(x) := \frac{x}{a + \sqrt{a^2 + x^2}}, \quad a = m + \frac{1}{2},$$

proved best of its form for all real orders ≥ 0 by Segura [2]; we use [3, Thm. 2], which bounds $I_\nu(x)/I_{\nu-1}(x)$ for $\nu \geq \frac{1}{2}$ by $x/(\nu - \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2})$; setting $\nu = m+1$ bounds $\rho_m = I_{m+1}/I_m$ by $x/(m + \frac{1}{2} + \sqrt{(m + \frac{3}{2})^2 + x^2}) \leq x/(a + \sqrt{a^2 + x^2})$ with $a = m + \frac{1}{2}$, which is the displayed bound (the last step uses $\sqrt{(m + \frac{3}{2})^2 + x^2} \geq \sqrt{a^2 + x^2}$). Substitute the recurrences, write $u = \rho_m \in (0, 1)$, $c = 1/x$, $S = u + 1/u$, $P = 1/u - u$; clearing the positive denominators, the claim becomes $\Delta_m > 0$ where the *exact factorization*

$$\frac{\Delta_m}{2m(m+1)} = (S - 3c)(P - (2m+1)c) + (2m+1)c^2$$

holds as an algebraic identity (expand and collect; the coefficients of $u^{\pm 2}$ are $\pm 2m(m+1)$, the linear-in- c terms are $-4m(m+1)c[(m-1)u + (m+2)/u]$, and $(m-1)u + (m+2)/u =$

$\frac{1}{2}[(2m+1)S + 3P]$, $1/u^2 - u^2 = PS$). The second factor is positive because U_m is the positive root of $1/w - w = (2m+1)/x$ (the calibration $1/U_m - U_m = 2a/x$) and $w \mapsto 1/w - w$ is strictly decreasing, so $\rho_m < U_m$ gives $P > (2m+1)c$. The first factor is positive because $\sqrt{a^2 + x^2} > a$ gives $U_m < x/(2a)$, so $u < x/(2m+1) \leq x/3$ and $S > 1/u > (2m+1)/x \geq 3c$. The third term is positive. Strictness is strict throughout. (This is the proof of [6], reproduced so that Theorem 18 is self-contained; the ordered-field implication is machine-checked in `PhiMonotone.lean`, modulo the two classical inputs carried as named hypotheses.) $\square$

B Audit table (the ghost ledger)

Each row: what was wrong, how it was caught, and the rule it produced. All entries are documented with transcripts in `docs/SURFACE-CLOSURE-NOTES.md`.

#	artifact	catch / rule produced
1	random TP_2 scan missed adversarial corners	targeted corner search; adversarial grids mandatory
2	verification cell with unbound parameter	explicit binding rule for inline prints
3	near- π scan at low precision, false violation	precision rule $\text{dps} \geq 2.2\beta + 40$ for $t \geq 2$, $\beta > 30$
4	stale-PDF claim misattributed between voices	provenance ledger: attribute to exact source
5	float64 CDF increments ± 11 (impossible)	positive-representation preference
6	$t=2$, $\beta=240$ residue 0.363 vs 0.317	boundary $t \geq 2$ included in the dps rule
7	I_1 series truncated at 60 terms, $z \leq 480$	cutoffs scale with argument; scaled asymptotics for $z > 35$; β -scaling checks
8	c_3 sign flip at $\beta = 120$ (cancellation)	alternating-sum rule $\text{dps} \geq 0.6\beta + 50$
9	$R = 0$ corner misplaced at $(1, 1)$	complete critical-point census required in prose too
10	one-saddle prefactor off by exactly 2	census rule; the factor 2 later proved an exact parity
11	I_1 lower bound dropped the negative tail	sign audits are a separate mandatory pass
12	fixed quadratic window at π was provably false	interrogate obligations before attempting them
13	Thm. A second proof: rearrangement misquoted	layer-cake arc proof substituted (this revision)
14	Lemma sign error: floor stated as upper bound	corrected; per-pair statements now carry signs explicitly
15	“exact” labels on verified inequalities	tricotomy re-audit of every label (this revision)

References

- [1] D. E. Amos, *Computation of modified Bessel functions and their ratios*, Math. Comp. **28** (1974), 239–251.

- [2] J. Segura, *Bounds for ratios of modified Bessel functions and associated Turán-type inequalities*, J. Math. Anal. Appl. **374** (2011), 516–528.
- [3] D. Ruiz-Antolín, J. Segura, *A new type of sharp bounds for ratios of modified Bessel functions*, J. Math. Anal. Appl. **443** (2016), 1232–1246.
- [4] NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov>.
- [5] L. Eriksson, *Recurrence–Amos proof of the unit-step order-monotonicity of $(\log I_\nu)'$* , ai.viXra:2607.0020 (2026).
- [6] L. Eriksson, *A weighted Turán-type monotonicity lemma for modified Bessel functions via the calibrated Amos bound*, ai.viXra:2607.0017 (2026).
- [7] L. Eriksson, *A Wronskian identity for antisymmetrized sine sums*, ai.viXra (2026), number pending.
- [8] L. Eriksson, *Parity barriers for decoupling inequalities*, ai.viXra:2607.0018 (2026).
- [9] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge Univ. Press, 1944.
- [10] S. Karlin, J. McGregor, *Coincidence probabilities*, Pacific J. Math. **9** (1959), 1141–1164.
- [11] K. V. Mardia, P. E. Jupp, *Directional Statistics*, Wiley, 2000.
- [12] S. Dharmadhikari, K. Joag-Dev, *Unimodality, Convexity, and Applications*, Academic Press, 1988.